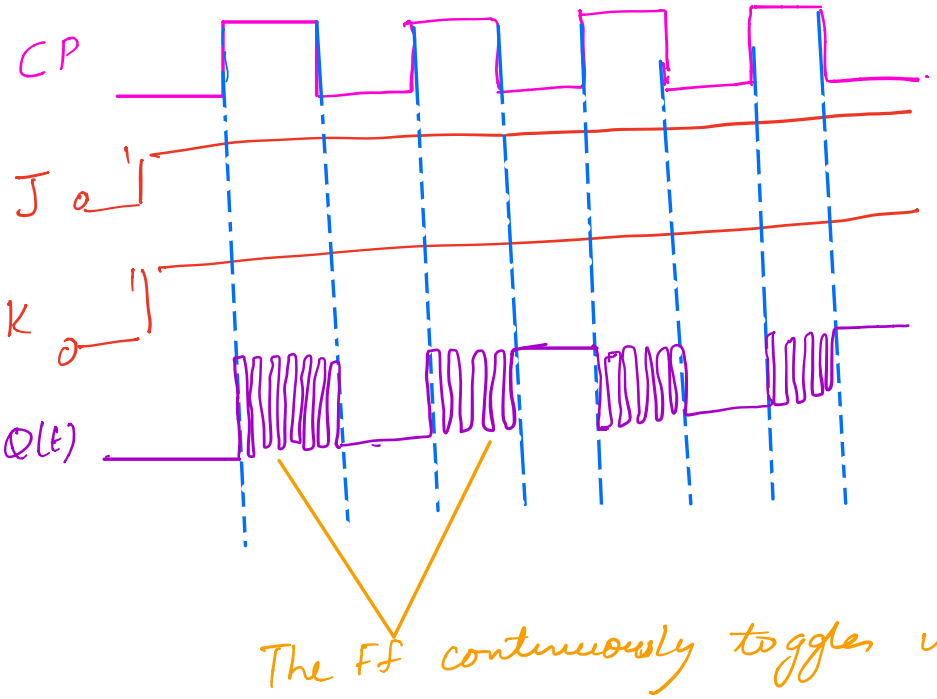
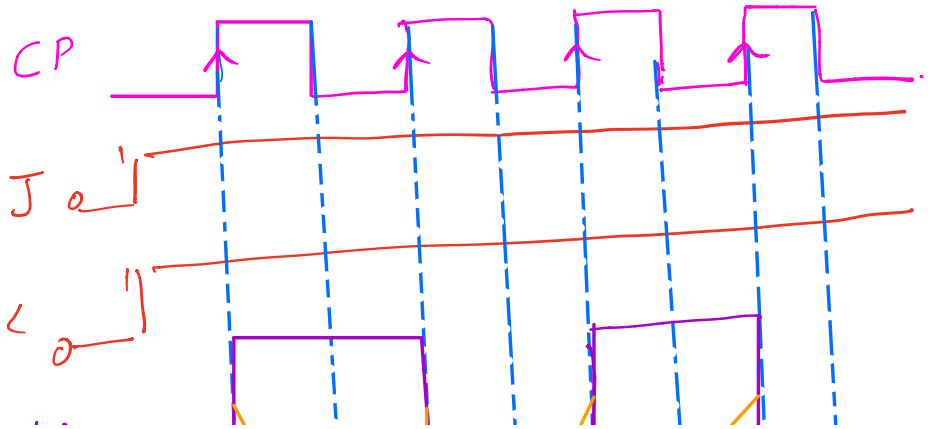
The Problem With Level Triggered FF:

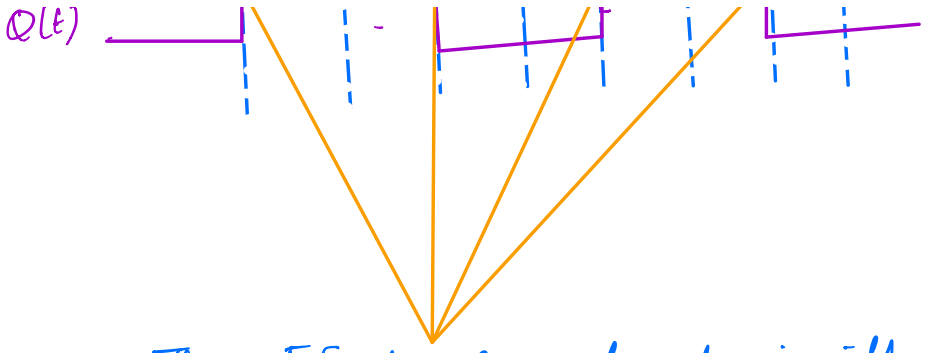
$Q(t+1) = \bar{Q}(t)$, when $J=K=1$ in JK FF, would mean that the o/p changes independent of any change in i/p. In a level triggered FF this results in



The FF continuously toggles when CP=1

We want

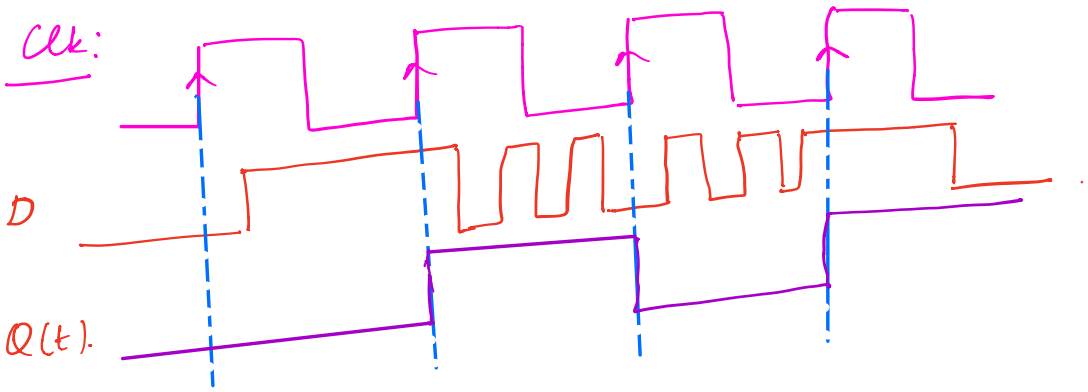




The FF toggles only during the rising edge of the CP or its is Rising Edge (+ve) Triggred FF.

So how does one construct or Design an Edge Triggered Ff ?

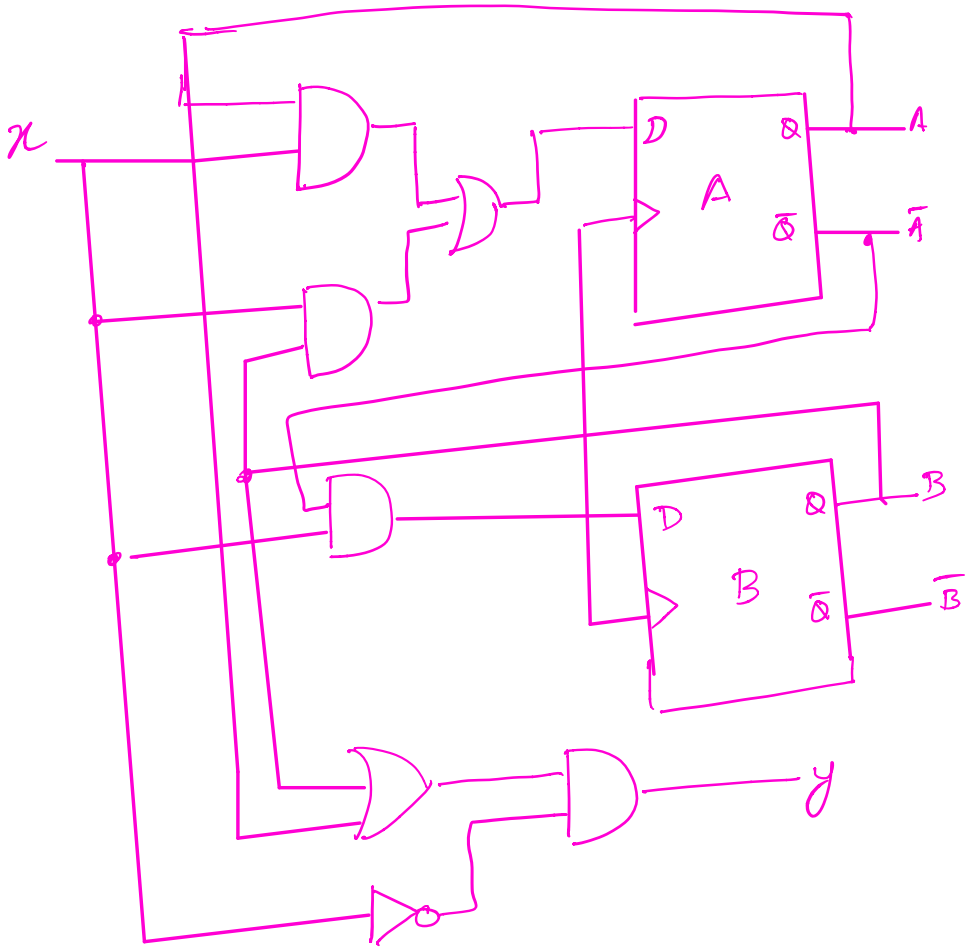
The Timing Diagram for the Edge Triggred D FF:



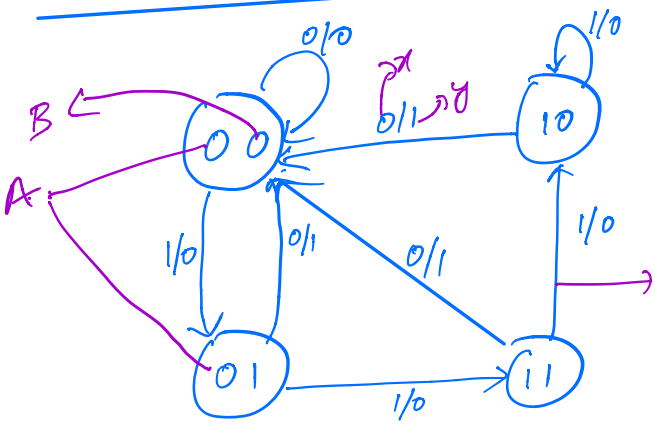
∴ Q changes only at the rising edge. Or Q(t) reflects D's value at the rising edge of the clock, Q (t+1) = D with the rising edge.

Changes to D between two rising edges does NOT affect Q.

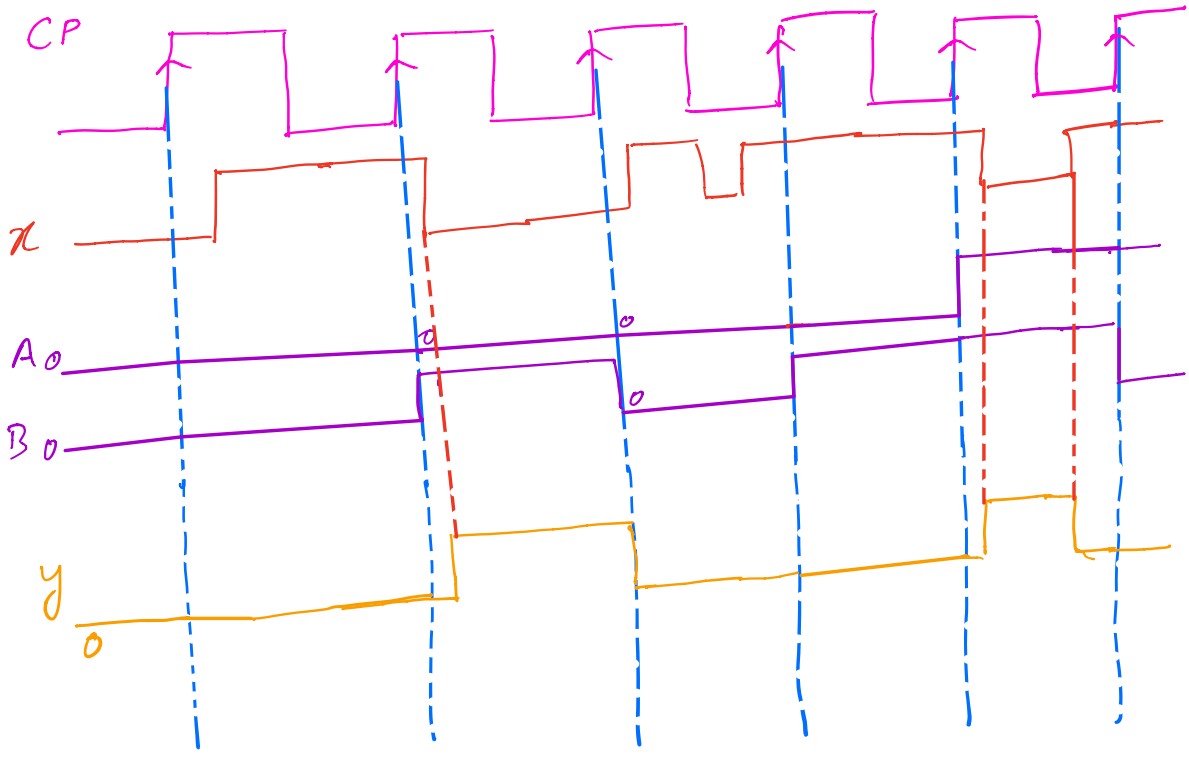
Analysis of Sequential Circuits



FSM for the above ckt



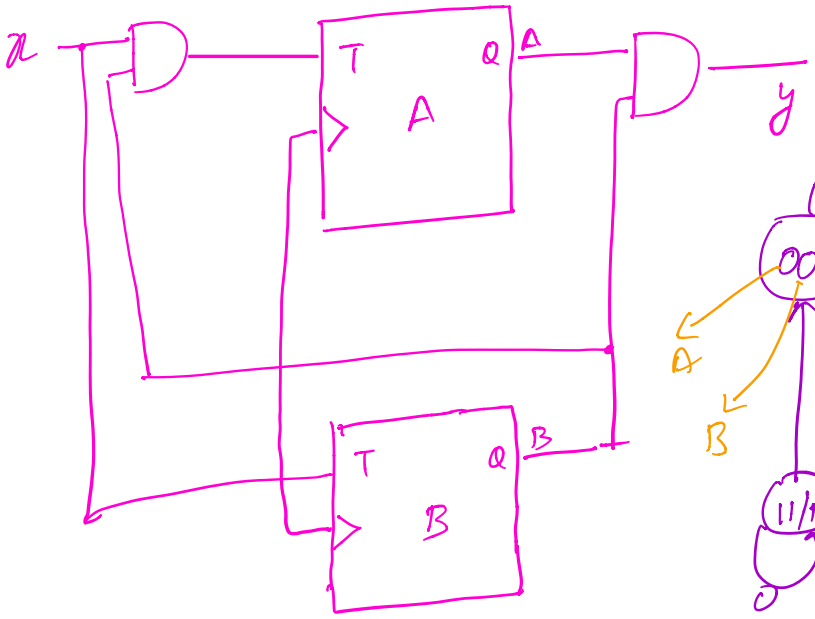
→ Transitions at the rising edge

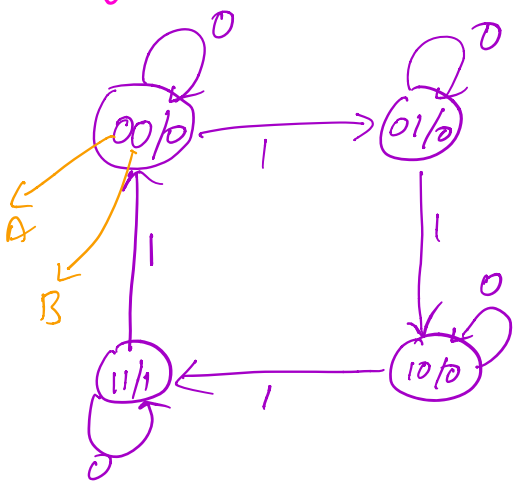


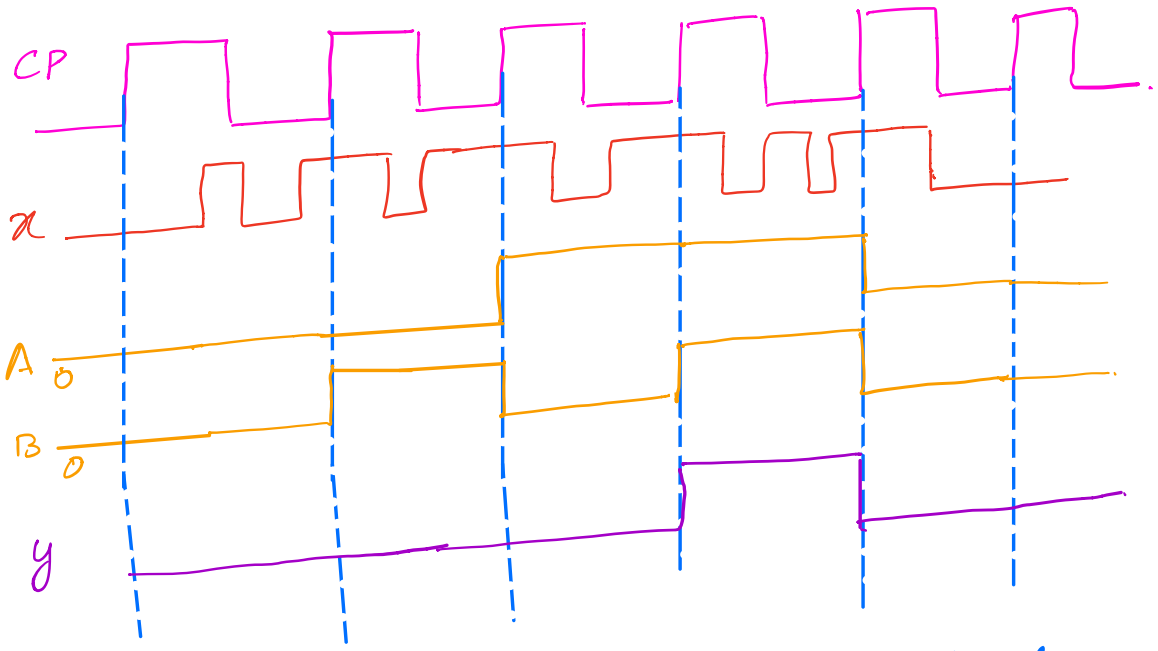
→ Changes in FF are synchronized with the CP.

→ Changes in combinatorial O/P y can occur when the FF changes at the rising edge OR when changes in i/p occur. Changes in y can be Asynchronous with the clock or MEALY FSM or Mealy Seq Ckt.

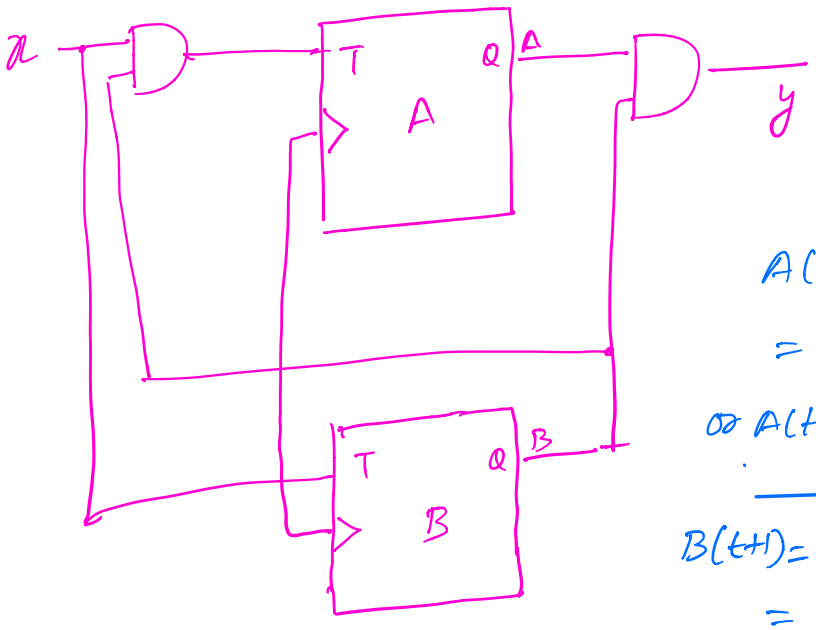
In contrast







MOORE FSM : Combinatorial o/p y is fn of only the FF states and changes in synch with the CP.



$T_A = Bx \longrightarrow (1)$

$T_B = x \longrightarrow (2)$

$y = AB \longrightarrow (3)$

$A(t+1) = T_A \bar{A} + \bar{T}_A A$

$= Bx\bar{A} + (\overline{Bx})A$

$\therefore A(t+1) = AB' + A'Bx + Ax' \longrightarrow (4)$

$B(t+1) = T_B B' + T_B' B$

$= x \oplus B \longrightarrow (5).$

Present State A(t)	Present State B(t)	I/p x	FF Input $T_A = Bx$	FF Input $T_B = x$	Next state A(t+1)	Next state B(t+1)	O/p y = A(t)B(t)
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	0	0	0	1	0
0	1	1	1	1	1	0	0
1	0	0	0	0	1	0	0
1	0	1	0	1	1	1	0
1	1	0	0	0	1	1	1
1	1	1	1	1	0	0	1